



FUEL TRANSPORTATION Department
SIMHADRI

Note No :1



Ref:SMTTP/FUEL TRANSP0/2024-25/ADM_APRV/645188

Date:

05/11/2024(11:11:41)

Subject:LMI on Guidelines for Signalling and Telecommunication system maintenance at Simhadri.

Subject:Administrative approval for LMI on Guidelines for Signalling and Telecommunication system maintenance

The LMI on Guidelines for Signalling and Telecommunication system maintenance, Ref no: LMI/OGN/OPS/SYST/061 is put up for approval of Competent authority.

Enclosures: 1. New LMI LMI/OGN/OPS/SYST/061 dated NOV 2024
2. OGN COS-ISO-00-OGN/OPS/SYST/058

Submitted by:

नरसिंहलु नीलपु - उप महाप्रबंधक

Narisimhulu Neelapu - 008049 (DY. GENERAL MANAGER ,FUEL TRANSPORTATION) ,Simhadri
(NNARISIMHULU@NTPC.CO.IN, 9491160670)

On:05/11/2024(11:11:41) **Ref:**SMTTP/FUEL TRANSP0/2024-25/ADM_APRV/645188

Note No:2

May be approved for implementation please.

Submitted by:

Amulya Kumar Behera - 097433 (ADDL. GENERAL MANAGER ,FUEL TRANSPORTATION)
(AMULYABEHERA@NTPC.CO.IN, 9425534368) ,Simhadri

On:05/11/2024(11:21:31) **Ref:**SMTTP/FUEL TRANSP0/2024-25/ADM_APRV/645188

Note No:3

May please be approved.

Submitted by:

आन्नप्रगद रमा कोटेश्वरराव - अपर महाप्रबंधक

Annapragada Rama Koteswararao - 047211 (ADDL. GENERAL MANAGER ,FUEL MANAGEMENT)
(ARKOTESWARARAO@NTPC.CO.IN, 8332988907) ,Simhadri

On:05/11/2024(11:31:05) **Ref:**SMTTP/FUEL TRANSP0/2024-25/ADM_APRV/645188

Note No:4

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)
(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri

On:06/11/2024(11:06:09) **Ref:**SMTTP/FUEL TRANSP0/2024-25/ADM_APRV/645188

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It is Rev. 0, to be mentioned along with LMI no. on every page

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

On:06/11/2024(11:55:49) **Ref:**SMTTP/FUEL TRANSP0/2024-25/ADM_APRV/645188

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Rev.0 mentioned along with LMI No. on every page, and modified LMI is here by attached.

Submitted for further processing.

Submitted by:

नरसिंहलु नीलपु - उप महाप्रबंधक

Narisimhulu Neelapu - 008049 (DY. GENERAL MANAGER ,FUEL TRANSPORTATION)
(NNARISIMHULU@NTPC.CO.IN, 9491160670) ,Simhadri

On:06/11/2024(12:24:49) **Ref:**SMTTP/FUEL TRANSP0/2024-25/ADM_APRV/645188

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Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

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Submitted by:

नरसिंहलु नीलपु - उप महाप्रबंधक

Narisimhulu Neelapu - 008049 (DY. GENERAL MANAGER ,FUEL TRANSPORTATION)
(NNARISIMHULU@NTPC.CO.IN, 9491160670) ,Simhadri

On:06/11/2024(15:31:32) **Ref:**SMTPP/FUEL TRANSPOR/2024-25/ADM_APRV/645188

Note No:9

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

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Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)
(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri

On:08/11/2024(10:15:24) **Ref:**SMTTP/FUEL TRANSP0/2024-25/ADM_APRV/645188

Note No:11

Submitted by:

तुषार कानूनगो - अपर महाप्रबंधक

Tushar Kanungo - 007981 (ADDL. GENERAL MANAGER ,SAFETY)
(TUSHARKANUNGO@NTPC.CO.IN, 9437960590) ,Simhadri

On:08/11/2024(11:17:10) **Ref:**SMTTP/FUEL TRANSP0/2024-25/ADM_APRV/645188

Note No:12

Submitted by:

सोमनाथ राउत - अपर महाप्रबंधक

Somanath Rout - 006476 (ADDL. GENERAL MANAGER ,IT)
(SOMANATHROUT@NTPC.CO.IN, 9437138223) ,Simhadri

On:08/11/2024(17:48:56) **Ref:**SMTTP/FUEL TRANSP0/2024-25/ADM_APRV/645188

Note No:13

Submitted by:

एस पद्मप्रिय - महाप्रबंधक

S Padmapriya - 004521 (GENERAL MANAGER ,TECH SERVICES)

(SPADMAPRIYA@NTPC.CO.IN, 9650992835) ,Simhadri

On:09/11/2024(12:07:27) **Ref:**SMTTP/FUEL TRANSP0/2024-25/ADM_APRV/645188

Note No:14

Submitted by:

प्रसन्ता कुमार जेना - महाप्रबंधक

Prasanta Kumar Jena - 006081 (GENERAL MANAGER ,O & M)

(PKJENA@NTPC.CO.IN, 9650990836) ,Simhadri

On:11/11/2024(20:01:58) **Ref:**SMTTP/FUEL TRANSP0/2024-25/ADM_APRV/645188

Note No:15

Approved & Submitted by:

संजय कुमार सिन्हा - मुख्य महाप्रबंधक

Sanjay Kumar Sinha - 005185 (CGM ,HEAD OF PROJECT)

(SKSINHA06@NTPC.CO.IN, 9431215239) ,Simhadri

On:12/11/2024(06:31:16) **Ref:**SMTTP/FUEL TRANSP0/2024-25/ADM_APRV/645188

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

Guidelines for Signalling and Telecommunication system maintenance

LMI/OGN/OPS/SYST/061 Rev.0

NOVEMBER-2024

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	Guidelines for Signaling and Telecommunication system maintenance
2	LMI No.	LMI/OGN/OPS/SYST/061 Rev.0
3	Reference Document	COS-ISO-00-OGN/SYST/061 Dt.Oct-2024
4	Superseded Document	NIL, New LMI
5	Prepared By	Name: N Narismhulu, Deputy General Manager (FT)
6	Checked By	Name: Amulya Kumar Behera, AGM (FC & FT)
7	Checked By	Name: A R Koteswara Rao, AGM (FM)
8	Checked By	Name: Navin Kishore, AGM (EEMG)
10	Checked By	Name: Tushar Kanungo, AGM (Safety)
9	Checked By	Name: Somanath Rout, AGM (IT)
11	Reviewed By	Name: S Padmapriya GM(TS)
12	Reviewed By	Name: P K Jena G M (O&M)
13	Approved By	Name: S K Sinha H O P (Simhadri)

INDEX

Sl. No.	Particulars	Page No.
1.0	Introduction	1
2.0	Superseded Documents	1
3.0	Details of Revision	1
4.0	Essential of signalling system	1
5.0	Maintenance plan	2-11
6.0	Special requirements of signaling in 25KV AC electrified area	11-12
7.0	Safety	13
8.0	Permit system	14
9.0	Records	14
10.0	References	14
11.0	Review	14
12.0	Distribution of LMI	14
13.0	Annexures	15-49

S No	Activity	Responsibility
1	Maintenance plan (5.0)	DGM(FT)
2	Wireless communication system (5.5.2)	AGM(IT)
3	Power supply (5.6)	DGM(FT)
4	AT Power (5.6.3)	DGM(FT)
5	Special requirements of signalling in 25 KV ac electrified area.	DGM(FT)

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

1.0 INTRODUCTION

Coal required for Thermal Power Plants are transported through railway system. Pit head plants have dedicated railway system known as MGR system and in rail fed plants coal is transported through Indian Railway.

Signalling & Telecommunication (S&T) system is used to control the movement of rakes in railway system. S&T system in railway has evolved with time and with evolution in technology. IR has established & standardized S&T system and at NTPC same system is implemented.

At broad level, S&T system has two purposes a) to move the rake safely and b) to utilize the asset (track) optimally. Railway system is capital intensive system hence both aspects are important.

Design feature of S&T system are a) Failed safe, b) Reliable c) Simple and d) Unambiguous. These aspects of S&T system need to be maintained at all times to ensure safe movement of Coal rakes and safety of working personals.

2.0 SUPERSEDED DOCUMENTS

Not Applicable

3.0 DETAILS OF REVISION

Not Applicable

4.0 ESSENTIAL OF SIGNALLING SYSTEM

4.1 General:

General use of signals: The railway signals mentioned here shall be used for controlling the movement of trains in all cases in which exceptions are not allowed by approved special instructions.

Some of the basic requirement which needs to be ensured are:

Fixed signals shall be clearly visible to the Loco Pilots of trains approaching them and shall be placed immediately to the left of or above the line to which they refer unless otherwise authorized by special instructions.

Every fixed signal shall be so constructed that, in case of failure of any part of its connections, it shall remain at, or return to its most restrictive aspect.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

Fixed signals, except automatic signals, shall always show their most restrictive aspect in their normal position.

The normal aspect of an Automatic Stop signal is 'Proceed'. Where, however, the signal ahead is manually operated, the aspect normally displayed may be 'Caution' or 'Attention'.

A Calling-on signal shall not be taken 'OFF' until the train has been brought to a stand at the Stop signal below which the Calling-on signal is provided.

Details of Basics of signalling system is given in Annexure-I

4.2 Interlocking

It may be defined as – a technique, achieved through the mechanical or electrical means by which it is ensured that before a signal is taking 'OFF', the route which the signal controls, is properly set, locked and held till such time the entire route is traversed by the Train and at the same time all the signals and points, the operation of which would lead to conflicting movements, are locked against the feasibility of such conflicting movement.

5.0 MAINTENANCE PLAN

5.1 General Guidelines:

A sample of the defects left in service should be pop marked to show their location and extent. The defects and pop marks should then be sketched or photographed. The depth of any grinding or regrinding should be accurately recorded. Repair to be taken up as per procedure recommended by OEM.

5.1.1. NTPC follows the Indian Railways system of preventive maintenance for maintenance of S&T system to achieve the maximum possible availability and reliability in service. The system of preventive maintenance envisages a schedule for maintenance attention at regular specified intervals. It aims to find potential failures before they fail in service due to ageing, wear, and tear, while also endeavoring to obtain maximum life possible for the components.

5.1.2. S&T system is fail-safe system by design i.e. whenever any apparatus fails it leads to safety. This design aspect is of paramount importance for the overall safety of system and to be maintained all the time.

5.1.3. HIRA & JSA document to be prepared for each maintenance activities and working personnel associated with S&T maintenance to be regularly briefed through pep-talk/TBT.

5.1.4. Some of the S&T equipment like axle counter, block instrument, signals belonging to station may lie on Indian Railway (IR) zone with their S&T system. Direct maintenance of these equipment may not be permitted to NTPC by IR. Maintenance of such equipment to be done as per IR private siding agreement and coordination with IR.

5.1.5. Seasonal preparation to be planned for summer and monsoon as per the guidelines issued by the COS-FT.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

5.2 Frequency of schedule maintenance:

Sr No	Equipment	Maint. Schedule Freq	Remarks
01	Track Circuit	Weekly	
02	Signals	Weekly / Monthly	
03	Point Machine	Weekly / Monthly /Half-Yearly	
04	LC Gate	Monthly	
06	Axle Counter	Monthly	
07	Relay Room	Monthly	
08	IPS	Monthly	
09	MSDAC	Monthly	
10	EI System	Monthly	
11	VHF System	Monthly	
12	DG Set	Monthly	
13	Apparatus Case /L Box	Monthly	
14	Earth Pit	Half Yearly	

Depending upon site conditions and equipment, the above schedule may be increased/decreased and/or other equipment may be added/deleted. Record of maintenance of all equipment to be maintained by station.

5.3 Tools for S&T Maintenance:

All personals engaged in the maintenance of signalling & telecommunication equipment shall be in possession of proper tools and adequate number of appropriate insulated tools like rubber gloves, pliers, screw drivers, etc. A rubber floor mat may be used for standing while attending to circuits carrying more than 110-Volt except in outdoor locations.

While using metal tools like screwdrivers, nose pliers, cutting pliers, etc., care shall be taken to avoid short circuiting and consequently malfunctioning of system / blowing out of the fuse.

Multi-strand wires shall be terminated on terminal with lugs and covered with insulation sleeves. While terminating wires, care shall be taken to bend the wires in clockwise direction.

List of tools is given at Annexure II

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

5.4 Signalling System:

5.4.1 Track Circuit

Track circuits are used to determine whether a particular portion of track is clear or occupied. A closed type of track circuit shall be provided to prove the clearance of rail track. Double rail track circuits shall be provided on non-RE areas, as far as practicable. Single rail or double rail type as convenient may be provided in RE areas. Also, the various portions of tracks in a track circuit shall, as far as possible, be connected in series & no portion of track to be bypassed.

Some of the basic requirement of track circuits which needs to be ensured are as follows -

1. Where continuous Track Circuits are installed, the polarity of the adjoining rails of the adjacent track circuits must be opposite, which is known as "staggered polarity".

2. FAIL SAFE ADJUSTMENT OF DC TRACK CIRCUIT

The circuit is properly energised during wet weather on minimum ballast resistance conditions and during dry weather on maximum ballast resistance conditions. The track circuit shall not be over-energised to such an extent that the shunting value drops below 0.5 ohms for all types of track circuits.

The following shall be kept in mind before adjusting DC track circuit.

1. Track voltage should not be more than 250% of its rated PV values.
2. TR should be excited at 125% of its rates pick value when RB (Ballast Resistance) is minimum and battery voltage is normal.
3. TR voltage should not be more than 85% of its DA value when 0.5ohm TSR is connected across the track circuit

Details of requirement of Track circuit & maintenance instruction are given in Annexure III. Maintenance instruction of Battery is given at Annexure IX

5.4.2 Signals

A "Signal" is a medium to convey a particular predetermined meaning in nonverbal form to Loco Driver. Every aspect of signal has pre-determined meaning like, red aspect meaning is stop dead, yellow aspect meaning is caution and green aspect meaning is proceed.

Maintenance instructions –

1. All terminals in CT racks, junction box, LED signal lighting unit, are tight and clean.
2. Connect only 110V AC supply to the units, as these are AC lit units.
3. Ensure that protective caps are replaced once terminations and selections are made.
4. Don't leave loose connectors or wires at LED Signal lighting units. This may cause false operations.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

5. Each signal unit shall be aligned correctly to the required distance and direction by employing the sighting hole along the side of the signal unit and by manipulating the separate horizontal and vertical adjustments.
6. Signal number plate shall be provided for each signal for clear identification.
7. Hoods of adequate size to counteract the effects of sunrays shall be provided. Where required, metallic mesh at suitable spacing shall be provided to protect lenses from outside interference.
8. All openings shall be gasketed and cable/wire entrance shall be sealed to make the enclosure watertight, dust-proof and vermin-proof,
9. The housing shall be kept locked to avoid unauthorized access. Detailed instructions for maintenance are given at Annexure – IV

5.4.3 Point Machine

Point Machine are used to operate turnouts electrically. This are critical for ensuring safety during movement of rail vehicles hence, utmost care to be taken during setting of it and any defect/abnormality to be reported to concerned control room immediately and rectified as soon as possible.

Some of the common aspect to be ensured are –

1. Condition of tongue rail
2. Proper setting of tongue rail with stock rail.
3. Proper oiling & greasing

Detailed instructions for maintenance are given at Annexure – V

5.4.4 Level Crossing Gate

All Level crossings (LC) must be provided with lifting gates, speed breakers, signals/blinkers, caution for road traffic, etc. (as per IR norms). Additionally, a separate arrangement to be kept for closing of gate in case of non-functional of primary gate. Preferably all such gate to be manned and a means of communication between the gateman & concerned control room to be provided.

Detailed instructions for maintenance are given at Annexure – VI

5.4.5 Axle Counter

Axle counter is a device for monitoring a specified section of track for the presence of vehicle. There are two types in use:

- Single Section Digital Axle Counter (SSDAC)
- Multiple Section Digital Axle Counter (MSDAC)

Single section axle counters are generally used to monitor a single-track section i.e there is only one entry point and one exit point for a particular track section.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

Multi Section Digital Axle Counter is used to monitor a portion of line called as “Track Section” confined by multiple Entry & Exit points. There are counting heads (Axle Detectors) at the beginning and end of each track section.

OEM instructions to be followed for detailed maintenance work. Detailed instructions for maintenance are given at Annexure – VII

5.4.6 Relay Room

Relay room is brain of signalling system. All the inputs are taken from field and interlocking are proved here before permitting any movement. Hence, to control access to relay room, it shall be always locked with two separate locks by maintenance & operation respectively and entry to be restricted to authorized personals only. Each entry to the relay room shall be recorded in register with date, time, reason & work done etc.

Following needs to be taken care in relay room -

1. At relay room, a relay index board shall be provided for locating relays easily.
2. It shall be ensured that as far as practicable, the relay room is not located in the vicinity of polluted area, loco shed or engine ash pits etc. In case it becomes unavoidable, necessary precautions shall be taken so that relays are least affected by the chemical fumes, dust etc. and are provided with dust free environment. Where there is concentration of relays, Air- conditioning shall preferably be provided.
3. In 'Q' style relay the code pins configurations for plug in relay shall not be changed by drilling additional holes on plug board to avoid fixing of a wrong relay which may lead to unsafe condition. In Siemen's K-50 relays, the Code pins position on base plate should not be changed and to be retained as fixed by the manufacturer.
4. All relay clips shall be checked for proper locking in the base when plugged in. Extreme care shall be exercised while removing the clips from the jacks of the plug-in type relay.
5. Relays involving external circuits shall have cross protection and double cutting. Signal control relays shall preferably be provided with double cutting.

Detailed instructions for maintenance are given at Annexure – VIII

5.4.7 IPS

The function of Integrated Power Supply (IPS) system is to provide a stable and reliable AC and DC power supply to the Railway signalling installations against all AC mains variations or even interruptions.

The SMPS based Integrated Power Supply (IPS) system is modular in design. It consists of the following modules:

- AC Distribution Panel (ACDP)
- SMPS based Float cum Boost Charger (FRBC) Panel
- DC Distribution Panel (DCDP)

IPS system shall be provided with Class B and Class C type two stage protection.

The IPS systems and its individual modules have earth terminals and these should be properly earthed to the IPS cabinet. The earth resistance shall not be more than 2 ohms.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

Earth provided shall preferably be maintenance free using earth resistance improvement material.

Battery Bank

The IPS system is suitable for charging 110 V battery bank of Low maintenance cells as per as per IRS S88/93 or VRLA Maintenance free cells as per IRS:S-93/96A. The purchaser shall specify the type of batteries to be used. The battery is to be installed in a separate room.

The float voltage of each rectifier module shall be set as given in the following table:

No of Cells	Auto Float Mode Voltage		Auto Boost Mode Voltage	
	VRLA Cells	LMLA	VRLA Cells	LMLA
55	123.8 V	118.25 V	126.5 V	133.5 V
1	2.25	2.15	2.3V	2.42V

Normal Float voltage for VRLA battery is 2.25 V/Cell and for conventional battery it shall be 2.15 V/ Cell and Normal Boost voltage for VRLA battery is 2.3 V/Cell and for conventional battery it shall be 2.42 V/Cell.

Battery under voltage isolation

The system has provision for battery isolation which shall be effective at

- (i) For VRLA battery: 1.8 V/cell (+0.012 V/cell)
- (ii) For low maintenance lead acid battery: 1.85 V/cell (+0.012 V/cell)

Battery under voltage adjustment is provided inside the switching control unit and it is adjustable from 1.80 to 2.0 V/cell. Battery shall get reconnected after restoration of mains.

These are typical values, refer to OEM guidelines for exact values.

The batteries shall be installed in a separate room/apparatus case to avoid damage to power equipment due to fumes. The battery room shall be provided with acid-proof flooring and acid- proof tiles on walls up to a suitable height. The battery room shall be well ventilated and provided with exhaust fans.

To check & ensure healthiness of battery bank, electrical power supply to IPS to be switched-off periodically (monthly or as specified by OEM) after ensuring that batteries are fully charged, put batteries on load & measure battery voltage, current & specific gravity till battery voltage falls to just above discharge value. The backup provided should not be less than 2/3 hours.

Detailed maintenance instructions are given in Annexure IX

5.4.8 EI System

Electronic Interlocking (EI) system is a microprocessor based interlocking equipment to read the yard and panel inputs; process them in a fail-safe manner as per selection table and generate required output.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

EI systems are very sensitive to power supply fluctuations hence necessary precautions to be taken as per OEM recommendations.

As EI system is OEM specific and parts from different

Following general precautions to be taken for EI system installation –

1. Maintain earth pit resistance within prescribed value. Earth pit for EI system installation should not be connected with Power system.
2. Maintain and record power supply.
3. Always stand on an approved conductive floor mat when touching or handling printed circuit boards.
4. Always wear a strap grounding device. The wrist strap should have a 1.0 megohm current limiting resistor. Connect the wrist strap grounding connector to suitable ground connection.
5. Keep the work area clean and free of debris. Avoid using non-conductive materials such as Styrofoam cups, plastic ashtrays, cellophane wrappers, or plastic covered binders in the vicinity of the cards and modules.
6. Maintain temperature of EI room and seal opening to avoid dust ingress. Refer OEM manual for detailed maintenance instruction.

(EI based signalling system is not available at NTPC-Simhadri)

5.4.9 Apparatus Case / L Box

Apparatus Case are used as Location Box, Battery Box, Junction Box etc. Maintenance of Location Box

1. Purpose of apparatus case and equipment installed inside to be lettered outside for easy identification.
2. Maintain cleanliness around Loc Box to give access in all weather conditions.
3. Apparatus case to be painted to avoid rusting.
4. All cable termination to be lettered properly.
5. There should not be any opening in Loc Box, as these will give entry to snakes/insects.
6. All apparatus cases should be earthed.
7. Loc Box must be properly locked to avoid unauthorized access and access of key to be controlled.

5.5 Communication System:

Clear & precise communication is important to ensure safety in MGR operation. Communication system for this purpose can be divided into two groups –

1. Wired Communication
2. Wireless Communication

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

5.5.1 Wired Communication

This type of communication system is provided between control rooms, LC Gates, maintenance staff, etc., for the SM/ASM to communicate where the communication media (path i.e. cable / OFC etc.) is owned & maintained by S&T. This may also be called fixed communications as the location of use is fixed.

Station should endeavor to provide wired communication at the location mentioned above to have direct and clear communication through Copper Cable, OFC, Radio-link.

5.5.2 Wireless Communication

This type of communication system is used to communicate with ground staff, Loco crew etc. who are moving. VHF, UHF, SIM Based Walkie-Talkies are used generally for this purpose.

Statutory compliance is required for the use of VHF equipment. The no of VHF equipment in use must be in accordance with license issued and license to be renewed periodically.

Station is required to maintain a record of issue & return of VHF equipment to user to keep track of it. Audit of VHF equipment to be done every six months to verify availability of all equipment, if any loss is found then a complaint to be lodged with local Police.

MGR operation specifically shunting & unloading activities requires frequent communication between parties to ensure safety. This communication is done through wireless devices (VHF/UHF etc.) which requires specific etiquette for effective & clear communication. To ensure the log of communication for training purposes & recording any miscommunication while giving instructions, voice logging equipment to be provided in the control room with storage backup of at least last 7 days.

Lightening Arrester (LA) with associated system must be installed to protect VHF antenna and its height must be more than VHF antenna.

How to use VHF equipment are given at Annexure - X

5.6 Power Supply:

At least two independent power supply sources to be available for supplying power to S&T system. If the control room is inside plant, then two sources from different sections of switchgear may be drawn. If it is outside plant, then one source from the state board and other from DG set to be provided.

5.6.1 DG Set

DG set may be provided at control rooms which are outside plant premises or where at least two different power supply sources are not available. DG set to be put into service for at least 30 min weekly to ensure healthiness of system and charging of starting battery. Record of running hours and consumption of fuel to be maintained by SM/ASM.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

5.6.2 Solar Power System

Solar power systems may be provided at Level Crossings and at control rooms as an additional source of power supply.

5.6.3 AT Power

If the track is electrified, then power supply may be drawn from it for supplying power to S&T System.

5.6 System Protection:

5.7.1 Fuses

Fuses shall be of an approved type. Fuse of correct capacity, which should be not less than 2.5 times the rated current, shall be provided. Following to be taken care -

1. Normally non-deteriorating type, 'G' type, alarm or cartridge type fuses shall be used.
2. Fuses shall be so grouped that blowing of a fuse has minimum repercussion on train operation.
3. The fuses when blown off, shall preferably give a visual indication.
4. Each group of circuits shall be carefully protected by fuses in the cabins and in location to facilitate easy fault localization.
5. Fuses shall be so arranged that they can be easily replaced without causing interference or unsafe conditions to other circuits.

5.7.2 Use of Multimeter

Multimeters are invariably used to measure electrical parameters. Proper setting of multimeter must be ensured before putting its leads to terminals. Any mal-setting may lead to damage to multimeter or to circuit leading to malfunctioning of S&T system.

5.7.3 Earthing

Earthing of electrical/electronic system is important for its safety and proper functioning. In addition to above earthing to be provided to body of electrical equipment for safety of working personals.

The resistance of earth for signalling circuits shall not exceed values given in annexure XI. If it is not possible to reduce the earth resistance below the specified value, additional earths pits may be provided in parallel.

Separate earth shall be provided for each block instrument at a station, Loc Box & Axle Counter, Electrical system & S&T system.

Where more than one earth is used, the distance between Earthing pits shall not be less than 3 metre. The conductor leading to these earthings shall be electrically insulated from each other throughout and also from metallic structures connected to different earths.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

As far as possible, the earthing arrangement shall be located in the natural soil. The made-up soil which has not consolidated or is likely to be eroded by weather, shall be avoided.

Resistance of earth pit to be measured half-yearly and record to be maintained. Date of testing and value of earth resistance to be written on pit chamber.

Other details are given at Annexure XI.

5.7 Spare Planning

To maintain the S&T system available all the time, minimum quantities of critical spares to be ensured all the time. Minimum critical spares to be maintained at station is given at annexure XII and specification is given at Annexure XIII.

6.0 SPECIAL REQUIREMENTS OF SIGNALLING IN 25 KV AC ELECTRIFIED AREA

The objectives of the special precautions taken with signalling equipment in AC electrified areas are:

- to ensure that interference from the traction system cannot, under any circumstances, cause the Signalling system to exhibit false clear indications, or in any other way imperil the safe operation of trains. This objective must be achieved even if the traction or Signalling system is in a faulty or anomalous condition.
- to ensure that signalling equipment continues to function normally in the presence of traction current.
- to minimize danger to life from electric shock derived from the traction system.

Signalling is affected due to following in AC Electrified area -

- Visibility of signals due to OHE structure
- Electrical clearances of signals due to live OHE close by
- Traction rail-return current
- Electrostatic and Electro-magnetic Induction

Visibility of Signals is taken care by the proper implantation of OHE structures during the design stage of OHE. Where electrical clearances are not possible to maintain, suitable protective shields for signal structures shall be provided.

The electrostatic induction is practically eliminated by transferring a circuit into underground cables protected by a metal sheath. The electromagnetic induction causes various currents and voltages to develop in conductors parallel to the track. These include the rails, traction return conductors where provided, cable sheath, any other conductors in the vicinity, and signalling and telecommunication circuits. The voltages that occur in the conductors appear as potential gradients. In addition, there are potential differences between various points along the conductors and the adjacent Earths. The value of this induced voltage depends on various factors, such as –

- Length of parallelism between the cable conductors and the electrified track.
- Soil conductivity.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

- c) Screening efficiency of the cable sheath where existing.
- d) Return current through the rails and return conductor where provided.
- e) Mutual inductance between catenary and the cable conductors.
- f) Separation between catenary and cable.
- g) The current carried by the catenary, etc.

In the matter of electrical clearances, the fundamental rule to be observed is that no one is allowed, under normal conditions, to approach closer than 2 meters from the extreme positions of the live parts of the O.H.E. The following protective measures shall, therefore, be adopted.

If any portion of a signal post or its fittings where signalling staff have to work, falls within 2 meters of a 25 KV live conductor, or any metal part electrically connected to this conductor such portion shall as far as practicable be protected by an iron screening of approved design solidly connected to the structural work.

If for any reason it is not practicable to provide the protective iron screening as given in para 22.2.6.1 above, a Caution Board of approved design shall be provided on the signal post at a height of 3 metres above the rail level, to caution the signal staff.

Note: Technical personnel shall exercise particular care to protect themselves while working on signal posts not provided with protective screens. If there is any likelihood of any part of their tools or equipment coming within 2 metres of live equipment, they shall take a power block as detailed in OGN: COS-ISO-00-OGN/OPS/SYST/058 "SAFE WORK PRACTICES IN ELECTRIFIED TRACK SECTION". The same precautions are also required in the vicinity of return conductors, which should be treated as live.

The cable shall be so laid that it is not less than one meter from the nearest edge of the mast supporting the catenary or any other live conductor, provided the depth of the cable does not exceed 0.5 meters. When the cable is laid at a depth greater than 0.5 meters, a minimum distance of 3 meters between the cable and the nearest edge of the O.H.E structure shall be maintained. If it is difficult to maintain these distances, the cable shall be laid in concrete / heavy duty HDPE/Ducts or any other approved means for a distance of 3 meters on either side of the Mast. When so laid, the distance between the cable and the mast may be reduced to 0.5 meters. These precautions are necessary to avoid damage to the cable in the event of the failure of an overhead insulator.

The length of D.C single rail track circuit in AC electrified areas with AC immune 9 ohms shelf type relay shall not exceed 450 meters where wooden sleepers are used and 350 meters where concrete sleepers are used. The length of track circuit can be extended upto 450 meters when QTA 2 plug-in relay is used.

In view of the increased AC immunity due to the presence of biased magnetic arrangement, QBAT relays can be used upto a maximum length of track circuit of 750 meters using one 'B' type choke at the relay end, under minimum ballast resistance of 2 ohms/km. Operation of track circuit with this type of relay will require four cells delivering 8.8V. QBAT relays shall be used in conjunction with QSPA1 relays conforming to BRS 933A.

For details OGN: COS-ISO-00-OGN/OPS/SYST/058 "SAFE WORK PRACTICES IN ELECTRIFIED TRACK SECTION" to be referred.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

7.0 SAFETY

Unsafe conditions at a workplace or site are hazards that have the potential to cause injury or loss of human lives and at the same time may incur damage to property and disrupt train traffic resulting in huge loss. Some of these hazards include erroneous safety procedures, malfunctioning equipment or tools, or failure to utilize necessary safety equipment such as goggles and masks. Unsafe conditions can be found in a variety of workplaces, for example Relay Room, Power equipment room, Level Crossing Gate, Station yard, Signal post, near point machine etc. in relation to S&T department. Hence it is necessary to ensure safety measures at workplace, both for maintenance personnel as well as equipment.

To ensure safety following to be ensured at workplace –

1. Always obey the safety rules.
2. Use safety equipment and follow precautions.
3. Following safe method and avoid short-cuts.
4. Always use proper tool for the job.
5. Do proper & time bound inspection & maintenance of equipment.
6. Ensure availability of safety equipment.
7. Ensure availability of firefighting equipment & first aid box.
8. Ensure proper training/ counselling of personals.

Safety During Maintenance

1. Plan the maintenance activity in advance and inform the control room.
2. Evaluate the hazards (HIRA) involved and prepare JSA accordingly.
3. For doing maintenance work during daytime in track circuit, point machine or any equipment infringing the rake movement on track, put red banner flag on both side of working site at
 - a. 100 Meter while working on straight portion in track (yard area – max speed 15kmph)
 - b. FM to FM while working in point/turnout (yard area – max speed 15 kmph)
 - c. 300 Meter while working in main line (max speed 70 kmph)
4. Red banner flag to be replaced with red colour light for doing maintenance activity after sunset.
5. Maintenance team must have walkie-talkie for proper communication and having idea of rake movement being done in yard.
6. Refer IOM from AGM (OS-FT), sub - “preventive measures/practice to be followed in S&T area”, dtd 06.06.2023 for maintaining safety of S&T equipment.

For preparing HIRA document OD “COS-ISO-00- OD/OPS/SAFT/001” may be referred.

Details of ensuring safety at workplace is given at Annexure – XIV.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

8.0 PERMIT SYSTEM

Whenever, in consequence of the line being under repair or for any other reason, special precautions are necessary, a Caution Order detailing the kilometre between which such precautions are necessary, the reasons for taking such precautions, and the speed at which a train shall travel, shall be handed to the Loco Pilot at the stopping station immediately short of the place where such precautions are necessary, or at such other stations and in such manner, as prescribed under special instructions.

Caution Order Format given at Annexure – XV.

9.0 RECORDS

A record to be maintained of maintenance activities for each equipment separately. The record should contain all the relevant parameters and activities done and it should be possible to track history of maintenance on equipment.

Details of record to be maintained is given at Annexure – XVI

10.0 REFERENCES

1. Indian Railway Signal Engineering Manual
2. Various maintenance booklets from CAMTECH, Gwalior
3. General & Subsidiary Rules of Indian Railway.

11.0 REVIEW

This LMI document shall be reviewed by the HOD-FC/FT on a 02 yearly basis or as necessary and the same shall be approved by HOP.

12.0 DISTRIBUTION OF LMI

Soft copy of the LMI shall be uploaded in the EEMG dept. website on Simhadri intranet LAN by the EEMG Department.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

13.0 ANNEXURE

Annexure – I Basic of Signalling System

Annexure – II List of Tools

Annexure – III Track Circuit

Annexure – IV Signals

Annexure – V Point Machine

Annexure – VI LC Gate

Annexure – VII Axle Counter

Annexure – VIII Relay Room

Annexure – IX IPS

Annexure – X How to use VHF Equipment

Annexure – XI Earthing

Annexure – XII Spares List

Annexure – XIII List of Specification for S&T Items

Annexure – XIV Safety

Annexure – XV Format for Caution Order

Annexure – XVI List of Records

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

Annexure I

Basic of Signalling System

1. Types of Signal

The signals used for controlling the movement of trains are –

- (a) fixed signals
- (b) hand signals
- (c) detonating signals and (d) flare signals.

2. Systems of working

(1) All trains working between stations shall be worked on one of the following systems, namely: —

- (a) the Absolute Block System
- (b) the Automatic Block System,
- (c) the Following Trains System
- (d) the Pilot Guard System,
- (e) the Train-staff and Ticket System
- (f) the One Train Only System.

Note - The Absolute Block and the Automatic Block Systems alone commonly used on every railway.

3. Essentials of the Absolute Block System

(1) Where trains are worked on the Absolute Block System,— (a) no train shall be allowed to leave a block station unless Line Clear has been received from the block station in advance, and (b) on double lines such Line Clear shall not be given unless the line is clear, not only up to the first Stop signal at the block station at which such Line Clear is given but also for an adequate distance beyond it; (c) on single lines such Line Clear shall not be given unless the line is clear of trains running in the same direction, not only up to the first Stop signal at the block station at which such Line Clear is given, but also for an adequate distance beyond it, and is clear of trains running in the direction towards the block station to which such Line Clear is given.

4. AUTOMATIC BLOCK SYSTEM

Essentials of the Automatic Block System on double line.— (1) Where trains on a double line are worked on the Automatic Block system,— (a) the line shall be provided with continuous track circuiting or axle counters, (b) the line between two adjacent block stations may, when required, be divided into a series of automatic block signalling sections, each of which is the portion of the running line between two consecutive Stop signals, and the entry into each of which is governed by a Stop signal, and (c) the track circuits or axle counters shall so control the Stop signal governing the entry into an automatic block signalling section that - (i) the signal shall not assume an `off` aspect unless the line is clear not only up to the next Stop signal in advance but also for an adequate distance beyond it, and (ii) the signal is automatically placed to `on` as soon as it is passed by the train.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

5. BLOCK WORKING

A. Means of granting or obtaining Line Clear - The running of every train shall, in its progress from one block station to another, be regulated by means of any one of or a combination of the following –

- (a) electrical block instruments of token or token less type,
- (b) track circuits (c) axle counters, or (d) electrical communication instruments.

B. Provision of instruments — Electrical communication instruments shall be provided at every station, except at class 'D' stations, where they may be provided under special instructions.

6. Placing of signals

Fixed signals shall be clearly visible to the Loco Pilots of trains approaching them and shall be placed immediately to the left of or above the line to which they refer unless otherwise authorized by special instructions.

7. Working of Signals and Points

Fixed signals generally -

- (1) Every fixed signal shall be so constructed that, in case of failure of any part of its connections, it shall remain at, or return to its most restrictive aspect.
- (2) A signal which has been taken 'off' for the passage of a train shall not be placed to 'on' until the whole of the train which it controls has passed it, except (a) in case of emergency to avert an accident,
- (3) No fixed signal shall be taken 'off' without the permission of the Station Master.

8. Normal aspect of signal

- (1) Unless otherwise authorized under approved special instructions, fixed signals, except automatic signals, shall always show their most restrictive aspect in their normal position.
- (2) The normal aspect of an Automatic Stop signal is 'Proceed'. Where, however, the signal ahead is manually operated, the aspect normally displayed may be 'Caution' or 'Attention'.

9. Conditions for taking 'off' Home signal

- (1) When a train is approaching a Home signal otherwise than at a terminal station, the signal shall not

be taken 'off' until the train has first been brought to a stand outside it, unless

- a) on a double line, the line is clear for an adequate distance beyond the Starter; or
- b) on a single line, line is clear for an adequate distance beyond the trailing points, or under approved special instructions for an adequate distance beyond the place at which the train is required to come to a stand.

- (2) Where a train has first been brought to a stand outside the Home signal, the signal

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

may be taken 'off', if-

- (a) on a double line, the line is clear upto the Starter, or
- (b) on a single line, the line is clear upto the trailing points or under approved special instructions up to the place at which the train is required to come to a stand.

(3) Except under approved special instructions, the adequate distance referred to in sub-rule (1) shall never be less than-

- (a) 180 metres at stations equipped with two-aspect lower quadrant or two-aspect colour light signals, or
- (b) 120 metres in the case of stations provided with multiple-aspect signals or modified lower quadrant signals.

(4) Where a sand hump of approved design, or under approved special instructions a derailing switch, has been provided for the line on which a train is to be received, they shall be deemed to be efficient substitutes for the adequate distance referred to in sub-rule (3).

10. Conditions for taking 'off' Outer signal

(1) When a train is approaching the Outer signal otherwise than at a terminal station, the signal shall not be taken 'off' until the train has first been brought to a stand outside the signal, unless the line on which the train is to be received in the station is clear— (a) in the case of a double line, up to the Starter signal, and (b) in the case of a single line, for an adequate distance beyond the first facing points.

(2) Where the train has first been brought to a stand outside the Outer signal, the signal shall not be taken 'off' unless the line is clear up to the first facing points, or upto the Home signal at a station where there are no facing points.

11. Conditions for taking 'off' Calling-on signal

A Calling-on signal shall not be taken 'off' until the train has been brought to a stand at the Stop signal below which the Calling-on signal is provided.

12. Use of fixed signals for shunting

(1) The Outer, Home and the last Stop signal of a station shall not be taken 'off' for shunting purposes.

(2) At stations where Advanced Starters are provided, Starters may be taken 'off' for shunting purposes, except where the interlocking interferes with this practice, in which case hand signals shall be used where Shunting signals are not provided.

13. Hand Signals

Exhibition of hand signals —

(1) All hand signals shall be exhibited by day, by showing a flag or hand and by night by showing a light as prescribed in these rules.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

(2) During day a flag or flags shall normally be used as hand signals. Hands shall be used in emergencies only when flags are not available.

Stop hand signal— Indication: Stop dead How given by day: By showing a red flag or by raising both arms with hands above the head as illustrated below:

Banner flags — A banner flag is a temporary fixed danger signal, consisting of a red cloth supported at each end on a post and stretched across the line to which it refers.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

Annexure II

Tools for S&T System

List of common tools in a toolbox of Signal Maintainer

S. No.	Description	Remarks
1.	Multimeter	
2.	Megger	
3.	Clamp Meter	
4.	Hydrometer	
5.	Cutting Plier	
6.	Nose Plier	
7.	Plier	
8.	Screwdriver (Set)	
9.	Wire stripper	
10.	Test gauge - for point testing	
11.	Tommy bar	
12.	Hammer	
13.	Ring Spanner Set	
14.	Open end spanner Set	
15.	Adjustable wrench	
16.	Box spanner with insulated handle for ARA terminals	
17.	TSR Meter	
18.	Search Light Torch	
19.	Soldering Iron	
20.	Earth Resistance Tester	
21.	Track Drilling Machine (for Track Circuit)	
22.	Blower	
23.	Vacuum Cleaner	
24.	Drill Machine with bits	
25.	Hacksaw	
26.	Measuring Tape, Metal Scale	
27.	Grease Gun	
28.	Allen key set	
29.	Tester	
30.	Cable Fault Locator	

In addition to the above some specific tools are required for maintenance of Axle Counter, IPS, EI system etc. Requirement of these tools to be accessed as per site requirements and made available.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

Annexure III

Track Circuit

A. TRACK CIRCUITS REQUIREMENT

1. Glued Joints should be preferred in place of Nylon Insulation Joints for defining boundary of track circuit and effort shall be made to replace Nylon Insulation Joints with Glued Insulated Joints.
2. Insulated Joints/Glued Joints wherever provided shall be maintained as square joints. Where staggering cannot be avoided the distance between staggered joints shall not exceed the minimum wheelbase of the vehicles.
3. Rail ends of insulated joints shall be square and true. All rough edges and burrs shall be removed from bolt holes. Battered ends shall be put right and the gap between the rails should be equal to the thickness of the end post.
4. Fish bolts at the joints must be kept tight and the sleepers well packed in the vicinity of the joints.
5. Proper drainage should be ensured so as to avoid flooding of tracks during rains, particularly in yards where watering of coaches is done and in water columns and ash pits.
6. Ballast shall be kept dean throughout the track-circuited section and care should be taken to see that the ballast is kept clear off the rails and rail fastenings. The clearance from the foot of the rail should not be less than 50mm. Well screened ballast right up to the formation level shall be provided.
7. Rail ends shall be kept free from brake-dust, dirt, sand, rust, other foreign materials etc. All rough edges and burrs at rail ends must be removed.
8. In case of turnouts and crossings, insulated stretchers, insulated gauge tie plates and insulated crossing plates shall be provided as per approved drawings.
9. GFN liners shall be provided in the track circuited area using concrete sleepers.
10. Track relays shall always be of metal to carbon contact type & normally be of plug-in type, unless otherwise specially permitted.
11. Track relays shall be located at the entering end of the track circuits wherever possible.
12. Where relays are likely to be subjected to vibration, they shall be mounted on shock absorbers such as rubber pad, foam etc.
13. Connections between the track relays and track repeating relays shall be made in accordance with the approved wiring diagram and must invariably incorporate both cross protection and double cutting arrangements.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

14. The feed and relay ends shall be connected by separate and individual cables.
15. Jumper connections shall normally be so made that the whole of track circuit is in series excluding traction return rail. When the rails of a track circuit are in parallel, care shall be taken that the jumper connections are effective.
16. Jumper connections shall be so arranged that they are as far as possible protected from damage. All connections to the track shall be of sufficient length and allow for rail creep. Connections shall be firmly fitted to the web of the rail.
17. The lead wires used for connecting the feed set to the feed end boot leg/track lead JB and the track relay to the relay end boot leg/track lead JB shall preferably be of copper conductor having a minimum cross section of 2.5 sq. mm.
18. Only 'J' type clip shall be used in glued joint portion of track.
19. Separate feed shall be provided for each track circuit.
20. Q series track relay must be used in conjunction with Q series slow to pick up relay as a repeater relay.
21. Wooden sleepers, concrete sleepers or any other approved type of insulated sleepers shall be provided for track circuiting. Concrete sleepers where used, shall have a minimum resistance of 500 ohms between insert-to-insert.
22. Rail screws should preferably be used in place of dog spikes at insulated joints.
23. In case of turnouts and crossings, insulated stretchers, insulated gauge tie plates and insulated crossing plates shall be provided as per approved drawings.
24. Jumper connections shall normally be so made that the whole of track circuit is in series excluding traction return rail. When the rails of a track circuit are in parallel, care shall be taken that the jumper connections are effective.
25. Jumper connections shall be so arranged that they are as far as possible protected from damage.
26. All connections to the track shall be of sufficient length and allow for rail creep. Connections shall be firmly fitted to the web of the rail.
27. For short jumpers, galvanized iron wires and for long jumper cables, may be used. Where galvanized iron jumper is used, duplicate jumpers shall be run. Where cables are used, the size shall not be smaller than 7/0.750 mm.
28. The lead wires used for connecting the feed set to the feed end boot leg/track lead JB and the track relay to the relay end boot leg/track lead JB shall preferably be of copper conductor having a minimum cross section of 2.5 sq. mm.
29. Bond wires/Rope wires - shall be duplicate installed close to the fish plate. Bond holes

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

shall be drilled with a twist drill and the bonds driven-in immediately. The wires may be fixed by using channel pins in the holes or by brazing/welding.

30. The resistance of rail and bonding per 1000 metres of track shall not exceed 0.5 ohm for track circuits longer than 700 metres. A rail and bond resistance upto 1.5 ohm per 1000 metres of track may be allowed for length of track circuit less than 700 metres.

31. For track circuited points or lines in a station, track circuit termination shall be provided sufficiently before the Fouling Mark so as to avoid infringement to the standard dimension by any portion of the vehicle. The distance between track circuit termination and Fouling Mark shall not be less than 3.0 metres.

B. Maximum length of track circuits

Maximum length of track circuit under different track parameter conditions shall not exceed the limits as given in the following table.

Maximum length of track circuit under different track parameter conditions shall not exceed the limits as given in the table below:

S. No	RE / NON-RE	Sleeper	Section (Yard /Block)	Min Ballast Resistance (Ohm/KM)	TSR (Ohm)	Max Length (Metre)	Type of Track Relay to be used
01	Non-RE	Wooden /PSC	Block	4	0.5	1000	QT Type 4 or Shelf type relay as per para 17.15.2
02	Non-RE	Wooden /PSC	Yard	2	0.5	670	- Do -
03	RE	Wooden	Block	4	0.5	450	QTA2 9-ohm AC Immune or Shelf type 9-ohm AC Immune
04	RE	Wooden	Yard	2	0.5	450	- Do -
05	RE	PSC	Block	4	0.5	450	- Do -
06	RE	PSC	Yard	2	0.5	350	- Do -
07	RE	PSC	Yard	2	0.5	750	QBAT in conjunction with QSPA1 with B type chock at relay end

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

Annexure IV

Signals:

Important parameters for LED Signals –

S. No.	Parameter	Main Signal	Calling-On Signal	Route Lighting Unit	Shunt Lighting Unit
01	Rated Voltage at I/p of current regulator	110 V AC $\pm 25\%$	110 V AC $\pm 20\%$	110 V AC $\pm 20\%$	110 V AC $\pm 20\%$
02	Current at rated voltage per unit at i/p of current regulator	140mA+10%, -20%	150mA+10%, -20%	25mA, $\pm 5\%$	50mA, $\pm 5\%$
03	Colour	Red, Yellow, Green	Yellow	Lunar White	Lunar White

Important Terms –

Blanking Mode - In blanking mode, a Main Signal Lighting Unit shall extinguish when input current drawn by the current regulator falls outside specified limits of rated input current or illumination falls to a value which is less than 40% of nominal illumination due to a failure or any other reason. In such case, current regulator should not draw input current more than 30 mA at maximum rated voltage.

Non-Blanking Mode - In non-blanking mode, a Main Signal Lighting Unit shall remain lit when input current drawn by the current regulator falls outside specified limits of rated input current or illumination falls to a value which is less than 40% of nominal illumination due to a failure or any other reason. In such case, input current drawn by current regulator shall be limited to less than 40 mA to ensure dropping of ECR. Limit on input current shall apply when illumination has deteriorated to a value, which is not less than 40% of nominal illumination.

Note - As per RDSO guidelines, for RED aspect, Non – Blanking mode should be selected in Current Regulator & for Green Aspect, the Blanking mode should be selected. For Yellow aspect, it should be selected based on installation.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

Annexure V

Point Machine

Electric Point Machine shall be provided with suitable gaskets and wire entrances shall be sealed to make them watertight and dust proof. All the extra openings shall be closed as far as possible. Electric Point Machine covers shall be locked. The cover provided for insertion of crank handle shall also be locked with separate key providing access only for crank handle and crank handle key. Proper drainage shall be provided to avoid water logging near point area.

Normally the machine shall be installed beside the close switch leading to high-speed movements clear of all infringements.

Ground fittings of approved design shall be used. Ground connections shall be thoroughly checked for any crack or improper welded joint.

The machine has been properly cleaned, greased and oiled in accordance with the manufacturer's instructions and is working freely.

The top of sleepers which are to support the point machine are at level and that surrounding ballast is well tamped to prevent sinkage during the passage of train.

Notches/Notch profile of the locking, driving and detection slides should not be tampered/reconditioned. Worn-out slides should be replaced with new ones.

The essential requirement for interlocking, condition of tongue rail & obstruction test of point to be periodically checked.

Obstruction Test: - The point driving rod and the lock connections of the machine must be so adjusted that with 5 mm thick test piece obstruction placed between the switch and the stock rail at 150 mm from the toe of the switch: -

- i) the point cannot be locked,
- ii) the point detector contacts should not assume the position indicating point closure &
- iii) Friction clutch should slip.

It shall be ensured that during obstruction in the point, the feed to point is automatically disconnected within 1.5 times to 2 times the normal operating time of point machine and reconnects only when a fresh operation is done. Where sequential operation of point machine is done, the maximum time before disconnection may exceed to 3.5 times the normal operating time of one machine. For the machines provided with overload current protection, the feed to motor shall be disconnected with current ranging between 1.5 to 2 times its normal operating current. It shall reconnect only when a fresh operation is done.

Machines shall be kept in good condition free from rust, dust and dirt.

All gearing and bearing shall be properly lubricated according to manufacturer's instructions.

Commutator shall be kept clean, smooth and have bright appearance. Commutator may be

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

cleaned with chamois leather Under no circumstances, shall emery paper be used.

Brushes shall be kept clean and properly bedded on the commutator. Brushes shall have proper pressure and shall be free in brush holders.

Test the spring in each switch and ensure sufficient spring. It shall not be too excessive to cause excessive wear on machine part, neither it shall too less to hold the switch against the stock rail. In either of the case, the point drive rod shall be adjusted.

Voltage and current shall be checked periodically at the motor terminals. This will indicate any undue friction on the points or improperly fastened terminals at cable terminations.

Point Controlling Circuits - Point shall be set to reverse position. After the point has been fully reversed, track circuit controlling the point shall be shunted and operation of points to normal position shall be attempted. It shall not be possible to operate the point under these conditions. These tests shall be repeated with the point set in normal position, attempt being made to operate it to the reverse position.

The out of correspondence test shall be carried out by opening cut out contact of one end of point machine and the point lever/knob/button operated. The other end of the point may operate but the point indication relay must not energise.

Point Indication Circuit

The point shall be operated from normal to reverse and reverse to normal and the position of point detection relay as well as the indication of the point in the cabin/panel shall be checked for correspondence with the position of the points at site.

It shall also be checked that with the obstruction in the point, the detection relay is de-energised and both normal and reverse point indication in the cabin/panel flash in case of PI /RRI installations.

The operation of the detection relay to the correct position as well as its de-energisation should be checked by making and breaking the relevant point detector contacts at site.

Crank handle interlocking - The insertion of hand crank should disconnect the power supply to the machine.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

Annexure VI

Level Crossings

COMMUNICATION: Telephone communication between the Gate-lodge and the Station Master's office shall be provided.

APPROACH WARNING: Audible warning by a bell/buzzer of approved type operated by the approach of a train shall be provided at interlocked gates.

A. MECHANICAL LIFTING BARRIERS

1. All moving parts shall be properly lubricated.
2. Gears, rope drums and wire rope shall be doused with soft grease.
3. Cleaning of rodding and wire transmission should be done properly. Cleaning of all pipes and ducts to prevent obstruction by accumulation of dirt shall be ensured.
4. All bolts and nuts shall be tightened, and split pins shall be split properly.
5. Worn out pins, if any, in the crank shall be replaced.
6. It shall be ensured that both the booms are parallel in open condition.
7. Proper adjustment of wire sag and tension shall be made.
8. Integrity of interlocking and boom locking shall be checked.
9. Rusty wires and rodding shall be replaced.
10. It shall be ensured that 'E' type lock which locks the winch should also lock winch handle in closed position of barrier.
11. All gate locking apparatus must be examined regularly and working parts kept oiled. Gate locks must be kept in good working order.
12. Proper working of audio/visual warning, if provided, shall be ensured.

B. ELECTRIC LIFTING BARRIER

1. Machines must be kept in good condition, free from rust, dust and dirt.
2. The bearing surface and moving parts of the electric lifting barrier shall be lubricated properly.
3. Contacts must be kept clean and in proper adjustment. If contact surfaces are pitted, they must be replaced.
4. Commutator and Brushes.
 - a. Commutator must be clean, smooth and have a bright appearance.
 - b. Brushes must be kept clean and properly bedded on the commutator. Brushes should have proper pressure and should be free in brush holders.
5. Friction clutch must be so adjusted that the same should get de-clutched when the boom is raised or lowered fully and the power supply is not cut off.
6. The shock absorber at the boom rest (stop post) must be kept properly adjusted.
7. The time of operation of electric lifting barrier shall not exceed stipulated time. Measured value of current should not exceed rated values.
8. The satisfactory functioning of the lifting barrier by a hand crank shall be checked.
9. All gate locking apparatus must be examined regularly and working parts kept oiled. Gate locks must be kept in good working order.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

10. Proper working of audio/visual warning shall be ensured.
11. Approach warning arrangement where provided, is working satisfactorily and effectiveness of Approach and Back locking provided shall be ensured.
12. The telephone communication facilities were provided between Level Crossing gates and the adjoining stations shall be checked in regard to their satisfactory functioning.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

Annexure VII

Axle Counter

A. SSDAC -

Axle counter consists of track device mounted on the rails, trackside electronic equipment provided near the track and connected with track device & evaluator. Evaluator monitors the counts of track device (s) to give clear/occupied indication. Two intelligent track side electronic equipment may also be connected together to give clear/occupied indication.

Axle counters may also be provided in lieu of conventional track circuits for block working, viz. Intermediate Block Signalling, Automatic Signalling, Block Working controlled by track circuits, etc.

The track detection unit shall be fixed minimum 6 sleepers away from the nearest rail joint.

The distance between the 2 rail inductors of different axle counting circuits shall be at least 5 metres so as to avoid mutual interference.

To protect track fittings against damage from hanging parts of moving train, protectors shall be mounted on both sides of the fitting on single line and one side on double line. Protectors shall be placed in the sleeper space next to the track fittings.

Cable for transmitter coils and the receiver coils shall be laid in different pipes at a minimum separation of 500 mm.

Maintenance of axle counters manufactured by other manufacturers shall be as per instructions given by the manufacturers.

Tie Tamper shall not be used for 4 sleepers on either side of the axle counters. Incidentally they should be manually maintained.

Following general instruction to be followed

- Ensure that Receiver and Transmitter coil cables have been laid in different pipes.
- Ensure that both the TX coils & Rx coils are having proper alignment on Rail.
- Resetting should be done only after ensuring that there is no train in the section.
- Check that metal sheaths of the outdoor cable are connected to earth at both ends.

For maintenance of Single Section Digital Axle Counter various parameters are to be checked periodically. The various signal input and output levels and its limits which are to be recorded and adjusted to correct levels wherever necessary as per OEM instruction.

B. MSDAC –

MSDAC works on high frequency and using 'Phase Modulation' with phase reversal of 180 degree out of phase unlike conventional axle counter which works on 'Amplitude Modulation'.

On Indian Railways Multi Section Digital Axle Counters are designed as per RDSO Specification No. RDSO/SPN/176/2005.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

Annexure VIII

Relay Room

GENERAL

In case of big yards and busy stations where a large number of trains/ shunting movements take place resulting in intensive relay operations, relay room shall be air conditioned, at way side stations where the traffic density is less, the provision of air conditioning is optional.

Relay rooms in areas which are prone to dust or in vicinity of chemical factories releasing injurious fumes or in areas with extreme temperature may be air conditioned.

Suitable standard earthing shall be provided for all operating panel, power supply, switch board, transformers, inverters etc.

It shall be ensured that such a storage place is not in the vicinity of damp or chemically polluted environment.

Sealing screws of the relays shall be intact.

A contact chart shall be prepared duly showing the contacts used circuit-wise with reference and spare contacts available, wire count for each contact and relay position in the rack.

Line Relays shall be energized at the rated voltage specified in the specification.

Rack to rack wiring shall be generally carried out with multi core cable having plain annealed copper single core conductor of size 1/0.6 mm or multi-strand 16/0.2 mm as per approved specification for indoor cables.

FUSES

Fuses shall be of an approved type. Normally non-deteriorating type, 'G' type, alarm or cartridge type fuses shall be used. When fuse is not provided with fuse blown of indication, additional indication circuit may be provided.

At the time of commissioning of any signalling installation, the normal load current of every circuit shall be measured and recorded. These recorded values shall be checked with the theoretically obtained values. Fuse of correct capacity which should be not less than 2.5 times the rated current, shall be provided.

When there is a case of fuse blowing off, the concerned circuit current shall be measured and compared with the original recorded value. If there is a variation, action shall be taken to locate and remove the defect before a new fuse is inserted.

Cleaning shall be done regularly so that dust is not deposited on the relays.

Anti-tilting arrangement for shelf type relays is available and relays shall not be tilted or turned over to close contacts and relay contacts shall not be bridged.

INSPECTION

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

During maintenance visually inspect the relays. During visual inspection, relay shall be taken out of service if any of the defects are noticed in respect of:-

- (a) Movement of armature and contact carriage.
- (b) Wiping of contacts.
- (c) Arcing of contacts.
- (d) Pitting or charring of contacts.
- (e) Dust accumulation on contacts.
- (f) Electro-plating.
- (g) Corrosion, rusting of components.
- (h) Crack or breakage in components.
- (i) Presence of fungus and ants inside the relay casing.
- (j) Charring of cover near contacts in the case of plug-in-type relays.
- (k) Corrosion of Label.
- (l) Absence or tempering of seal.
- (m) Any other abnormal condition.

CABLES

Cable Laying

1. Adequate spare conductors to a minimum of 20% of the total conductors used shall be provided for in each main cable up to the farthest point zone, beyond this there should be a minimum of 10% spare conductors of the total conductors used. No spare conductors are required if the total number of conductors used is 3 or less. The spare conductors shall be provided on the outermost layer.
2. As far as possible low-lying areas, platform, drainages, hutments, rocky terrains, points and crossings, shall be avoided. Proximity to Electrical Sub Stations/Switching stations shall be avoided.
3. Provision of suitable duct/pipe/other arrangements shall be provided for bridge for laying of cables.
4. For drawing out cables, the cable drums shall be mounted on cable wheels or horizontal revolving platform. It should be ensured that no kink is formed while drawing out the cable. Drawing out of cable should be done by rotating the cable drum and not by pulling the cable with excessive force.

Trenching

1. The width of cable trenches should be commensurate with number of cables. The minimum width shall be kept as 0.3 meters. The bottom of the cable trench should be levelled and got rid of any sharp materials. In the soft ground, the cable should be laid at the bottom of the trench previously levelled. In the rocky ground, the cable should be laid on a layer of sand or sifted earth of 0.05 meters thickness previously deposited at the bottom of the trench. In both the above cases, the cable should be covered with a layer of sand or sifted earth of 0.10 meters thickness and hereafter a protective cover of trough or a layer of bricks or split DWC-HDPE pipe of approved type or any other approved

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

means should be placed.

2. Bottom of Trench shall be cleared in such a way, so as to remove all the boulders having sharp edges. In rocky soil, the bottom of the trench shall be filled with river sand or any other soft material and cable shall be laid on smooth surface.
3. The main cable laid parallel to the track shall normally be buried at a depth of 1.0 meters from top of cables to ground level, including those laid across the track (1.0 meter below the rail flanges). The depth shall not be less than 0.50 meters for tail cables. In theft prone areas, the cables may be laid at a depth of 1.2 meters with anchoring at every 10 meters.
4. After laying the cables, back filling of the trenches should be done properly. The soil excavated shall be put back on the trench, rammed and consolidated.
5. Cable markers of approved type shall be provided which should be placed at suitable interval and at diversion points to mark the route.
6. Where the cables are entering the Cable room/Relay room/apparatus case etc., the cables shall be firmly held with suitable clamping arrangement, so as to avoid downward displacement. Suitable ducts, ladder supports, cable guides etc., shall be provided for the cables inside the Cable room/Relay room. Sharp cable bends shall be avoided.
7. When a cable has to cross the track, it should be ensured that –
 - a. The cable crosses the track at right angles.
 - b. The cable does not cross the track under points and crossings.
 - c. The cable is laid in concrete/GI/DWC-HDPE pipes of approved type, suitable ducts or in any other approved manner while crossing the track.
 - d. Wherever feasible horizontal directional drilling method may be used.

Testing

1. The insulation resistance tests should be made when conductors, cables and insulated parts are clean and dry.
2. A 500 V insulation tester should be used for insulation testing of signalling cables and 100 V insulation tester shall be used for Quad cables.
3. Any metallic sheath or metal work of any rack or apparatus case should be bonded to earth during test.
4. Disconnect all cores of a cable at both ends. The disconnection may be made through links of ARA/Modular terminals, if provided.
5. Connect one terminal of the insulation tester to the conductor under test and other terminal to all the other conductors being bunched together and connected to earth. Similarly test remaining conductors of the cable one by one as in mentioned above.
6. Insulation Resistance so measured should not be less than 5 mega ohms per km at buried temperature, if the insulation resistance is found to be lower than 5 mega ohms, the cause should be investigated, and immediate steps taken to repair or replace it.
7. Insulation Resistance tests should be made in such a manner that safe operation of trains is not affected. It should be ensured that no unsafe conditions are set up by the application of test equipment.
8. All conductors in signalling cables must be tested for their insulation before commissioning.
9. The conductors of the cables may accumulate electrostatic charge. The cable conductors

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

should be shorted or earthed to completely discharge any accumulated charge.

- a. Before connecting the insulation tester while commencing the test.
 - b. After the insulation tester is disconnected when the test is completed. This is in the interest of safety of personnel and protection of equipment.
10. The results of the insulation resistance tests should be recorded in approved proforma. A comparison of test results between successive tests carried out on a cable under similar conditions will give an indication of the trend towards deterioration of the cable.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

Annexure IX

IPS System

Standard Configurations

RDSO has standardized five configurations for SMPS based IPS according to the type of station to cater for the power supply requirements of signalling gears.

Date of Commissioning of IPS System & Battery Bank to be displayed at IPS rooms.

Maintenance Check Points for ACDP

S. No.	Check Point	Action Required	Frequency
01	O/P voltages of Inverters, AVR's, Step Down Transformers as per requirement	Check if the O/P voltages are set as per requirement. If not, check for fuses contactors.	Once in 15 days
02	Inverter O/P current	Check if the DPM is showing the signal current as per original settings.	Once in 15 days
03	Auto changeover between Inverters and Bypass CVT	Check for this changeover operation as described in instruction manual	Once in a month
04	Connector Mechanical Alignment	Check if all the connectors of various sub systems are inserted properly	Once in a month
05	Sub System Cleaning	Remove one by one and clean them using a soft cloth. Gently blow some air to remove the dust inside.	Once in 3 months

SMR Maintenance

SMRs are fully alarmed and operate in an active loop sharing arrangement. However, some regular checks can be an early warning of problems waiting to happen. Check for failure signal by switching off one SMR at a time once in a month.

Current sharing (when number of modules is more than one number)

Under normal conditions the currents contributed by every rectifier should be within +/-2 Amps of each other

Maintenance Check Points for DC-DC Converters

S. No.	Check Point	Action Required	Frequency
01	Converter O/P voltage	Check if the O/P voltages are set as per requirement. If not, correct them	Once in 15 days
02	Paralleling of Converters	Switch OFF main Converter and observe if the stand-by is taking the load.	Once in a month
03	Paralleling of Converters	Switch OFF Stand-by Converter and observe if the main is taking the load.	Once in a month

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

04	Converter Mechanical Alignment	Check if all the converters are inserted properly	Once in 3 months
05	Converter Cleaning	Remove one by one and clean the converter using a soft cloth . Gently blow some air from top or Bottom to remove the dust inside.	Once in 3 months
06	Converter failure	Check for failure Signal	Once in 3 months

Adjustment during system working (without disturbing loads)

- Pull DC-DC Converter from front for adjustment.
- Connect the Converter to the rack with both I/P and O/P card extenders.
- Connect the test points to Digital Voltmeter.
- Adjust the potentiometer to get the desired output.
- Check all the converters voltages pertaining to individual applications to be equal.

Cleanliness

Apart from these instructions regular cleaning of IPS and battery room should be done. Cleaning of dust collected inside the IPS panels (SMPS, DCDP & ACDP) should be done with the help of blower once in a month.

Checking of earth resistance

Check earth resistance every 6 months with earth resistance meter. it should be less than 2 Ohm.

Battery maintenance

- For maintenance of Battery, manufacturer's maintenance manual to be followed.
- Some important points for battery maintenance are given below:
 - Maintenance
 - The temp. and Sp. Gravity readings of electrolyte have to be taken weekly.
 - Check float charge voltage and adjust to the specified value if required.
 - Check the electrolyte level and top up with distilled water whenever required.
 - Check the specific gravity of all cells in every month during the battery charge and the specific gravity shall be corrected accordingly.
 - If the specific gravity is higher than the specified, then it can be adjusted to the specified value by adding distilled water. Conversely if the sp.gr. is lower than the specified then add acid of 1.400 sp.gr
 - Ensure that specific gravity of electrolyte is uniform for all the cells & in the range of 1.200 ± 0.005 at 27°C .
 - Recharge the battery immediately after every discharge.
 - Check tightness of all electrical connections once a month.
 - Maintain proper records of charge, discharge, Cell Voltages, Sp.gr. etc.
 - Clean & wipe cells to remove acid, which may have dripped on cell during filling.
- **VRLA battery**
- Cleaning of all cells near its terminals.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

- Reading of all cell's voltage with Charger ON.
 - Reading of all cell's voltage with Charger switched OFF.
 - Boosting of Sick cell using Sick cell Charger.
 - Replacement of the faulty cell with new cell.
- **Low Maintenance Battery**
 - Cleaning of all cells around top cover and terminals periodically.
 - Applying petroleum jelly over the terminals.
 - Periodically recording of all cell voltages and their specific gravity.
 - Periodically checking of electrolytic level of cells.
 - Periodically recording of all cell voltages with Battery Charger switched OFF.
 - Boosting of Sick cell using Sick cell Charger.
 - Replacement of non-reparable cell in a bank by fresh charged cell.
- **Do's**
 - Always wear rubber gloves, aprons and goggles while handling acid
 - Keep the batteries & battery room area clean & dry.
 - Check & ensure proper polarity of the cells before connection.
 - Apply petroleum jelly to the connector portion (terminal bolt, nut & washers) for preventing corrosion of terminals.
 - Check the float voltage & current regularly and adjust to the specified value.
 - Provide adequate ventilation & illumination in the battery room.
 - Records should be regularly maintained.
- **Don'ts**
 - Do not mix batteries of different types or makes.
 - Do not make tap connections from Battery Bank.
 - Do not use metallic vessels for electrolyte preparation & filling.
 - Do not short circuit cells while using spanner, L-handle etc.
 - Never allow an open flame, spark or smoking in the battery room.
 - Do not discharge battery beyond DOD level.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

Annexure X:**VHF Equipment**

**How to Use VHF Equipment
Dos and Don'ts**

VHF equipment (Walkie-Talkie & Base Station) are one way radio i.e. at a time only one person can Talk and all other have to Listen. It is just like the single line operation for movement of Train where at a time only one Train is allowed to occupy the track section.

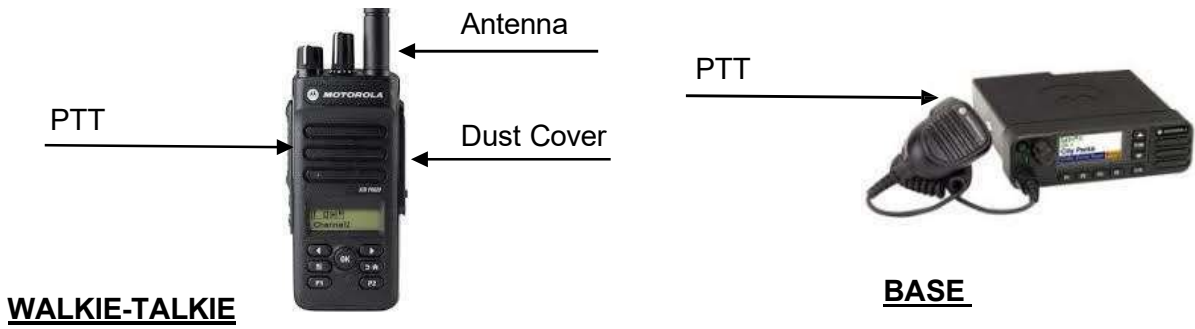
Do's

1. Ensure that your VHF equipment is tuned to proper Channel.
2. Ensure that channel is free before pressing the PTT button (starting communication) i.e. others are not communicating. If someone is communicating, please wait till their communication is completed.
3. Ensure that PTT button is pressed and hold before start speaking and keep it pressed while speaking, release it only after you have stopped speaking.
4. ALWAYS speak the name of person/Control Station first to whom you want to communicate with. Also, if you have received some message/communication ALWAYS acknowledge it.
5. Keep your message (communication) short i.e. break your message in short sentences in place of one long sentence.
6. Wait for some time before repeating message again, so that other can respond. Give a little pause between two communications.
7. When shutting down Loco, switch off Base Station too.
8. Give priority to placement activity at Track Hopper. If it is in progress, wait as far as possible, to completed it.
9. Ensure that Walkie-Talkie is sufficiently charged before issuing it. Enter it in register provided in Control Room and before leaving from duty return it and cancel it from register.

Don'ts

1. Do not press PTT button while someone is communicating, as it will disturb communication in process, even if you have not spoken.
2. Do not keep more than one equipment ON around you. It will echo (ring) your voice while you are talking on VHF.
3. Do not respond in hurry even if you have got the message. Let the first person complete his communication, pause a bit and then respond.
4. Do not hold/bend/twist the Antenna of Walkie-Talkie, it will damage it.
5. Do not remove the dust cover of programming port of Walkie-Talkie.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0



TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

Annexure XI:

Earthing

Earthing of cables, equipment, buildings, and structures is done for one or more of the following purposes:

- a) To ensure effective and rapid operation of protective equipment in the event of earth fault currents in an electrical supply system which might otherwise cause fire; and to protect against danger to life through shock due to non-current carrying equipment and metal work being maintained at a dangerous potential relative to earth.
 - b) To provide an earth for telephone systems employing an earth return signalling circuit.
 - c) To provide a means for the earthing of metal screens of telecommunications cables and equipment for the purpose of reducing interference.
 - d) To provide a direct connection with earth for discharge tubes, lighting protectors, etc.
 - e) To provide an earth to block instrument working on earth return.
- The earthing shall be provided at every location box where cables terminate.
 - It is not necessary to earth the sheath and armour of screened cables or armour of unscreened cables when they are used as a tail cable except in special cases where the length of the tail cable exceeds normal prescribed limits.
 - In case of signals falling within 2 meters from the electrified track, the protection screen shall be connected to an earth.
 - Earth wires shall be protected against mechanical damage and possibility of corrosion particularly at the point of connection of earth electrode.
 - The earthing lead shall be mild steel flat of size 35mm x 6mm or copper wire of 29 sq. mm cross sectional area (19 strands of 1.4 mm dia). In case the conductor is buried underground, it shall be protected from corrosion by an application of suitable anti-corrosive paint or bitumen or varnish. The length of the cable so treated shall extend half a metre beyond the buried length.
 - The earthing lead shall be soldered or crimped on a lug, which shall be bolted to the earth electrode. The nut & bolt to be painted with anti-corrosive paint.
 - The site for earthing shall be chosen in the following order of preference.
 - Where soil conductivity is poor, the chemical treatment may be resorted to improve the same. Common salt together with charcoal in alternate layers is generally used for this purpose and the addition of less than one part by weight of salt to 200 of soil mass may reduce the resistivity by 80% but there is little advantage in increasing the salt content above 3%. Calcium chloride and sodium carbonate are also beneficial.
 - Concrete cover of about 50mm thickness around the Earth for a radius of 1 metre to retain the soil moisture is desirable.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

- Under ordinary soil conditions, use of galvanized iron or mild steel electrode is recommended. In cases where soil corrosion is likely to be excessive, it is preferable to use either copper or copper clad electrode. The electrodes shall be free from paint, enamel or grease.
- Earths shall be regularly watered.
- All Earth connections shall be carefully examined and kept intact, and joints soldered. The wire between each earth and the connected equipment shall be electrically isolated.
- There shall not be any possibility of simultaneous human contact with metallic bodies connected to different earths, where it is not possible to provide suitable spacing or partition between various metallic objects referred to above, they shall be connected to a common earth.
- **Limits of earth resistance:** The maximum earth resistance specified is:
 - a) Earth for lightening discharger 10 ohms
 - b) Earth for equipment 10 ohms
 - c) Axle counter cable screen in ac electrified area 1 ohms
- Pressure of Return Current in Rail
- The flow of return current in the rails will give rise to a potential difference: -
 - a) between adjacent rails at an insulated joint of a track circuit or at an ordinary joint in case the fish plates and bonding are broken.
 - b) between the ends of a fractured rail at the fracture.
 - c) between an insulated rail and the non-insulated rail used for the traction return current; &
 - d) between the rail and the surrounding mass of earth.
- Wherever staff has to work on installations, which are in direct or indirect contact with the rails, they shall use tools (insulated and non-insulated) in accordance with approved instructions.
- Before any work is undertaken on Signalling or Railway Telecommunication cables, the staff shall take the following precautionary measures: -
 - a) Reduce the length of the circuit as much as possible.
 - b) Use Rubber gloves as far as possible or alternatively use insulated rubber mats; &
 - c) Before cutting the armour or the metal sheath of the cable or the wires in the cables, an electrical connection of low resistance shall be established between the two parts of the armour, the Sheathing and wires that are to be separated by cutting.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

Annexure XII:**S&T Critical spares list:**

S. N.	Type of Mat	Mat Description	Mat Code	Min. Stock at site/ central stores at any time	Remarks
1	Signal LEDs	Red	M7530116038	Higher of 10% population or 5 nos.	
2		Yellow	M7530116040	Higher of 10% population or 5 nos.	
3		Green	M7530116039	Higher of 10% population or 5 nos.	
4		Calling-On	M7530116037	Higher of 10% population or 5 nos.	
5		Shunt	M7530116044	Higher of 10% population or 10 nos.	
6		Route Jun (lunar)	M7530116041	Higher of 10% population or 10 nos.	
7		Route 1-9 (multi lamp or theatre type)	M7530990021	Min 2 nos.	Only If available
8		Route 1-19	M7530990020	Min 1 no.	Only If available
9	Track Circuit	T/F Charger	M7530156041	Higher of 10% population or 20 nos.	
10		Choke B type	M7530156040	Higher of 10% population or 10 nos.	
11		Track Resistance	M7530156038	Higher of 20% population or 20 nos.	
12		Nylon Block Insulations 52 Kg	M7530156005	Higher of 25% population or 20 set	
13		Nylon Block Insulations 60 Kg	M7530156002	Higher of 25% population or 20 set	
14		Glued Joint 52 Kg	M7530156009	Higher of 10% population or 10 nos.	
15		Glued Joint 60 Kg	M7530156010	Higher of 10% population or 10 nos.	
16	Relays	QSPA1 (TPR)	M7530116161	10% of population	
17		QTA2 (TR AC Immunized)	M7530116210	10% of population	
18		QT2 (TR)	M7530116207	10% of population	
19		Flasher	M7530116262	10% of population	
20		Timer	M7530116226	10% of population	

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

21		QL1 11F/4B	M7530116164	10% of population	
22		QN1 12F/4B	M7530116134	10% of population	
23		QN1 8F/4B	M7530116135	10% of population	
24		QNA1 8F/8B	M7530116152	10% of population	
25		QECX61 (ECR)	M7530116293	10% of population	
26	Cables	1.5x6C	M7530506011	1000 Mtr	
27		1.5x8C	M7530506012	1000 Mtr	
28		1.5x12C	M7530506014	1000 Mtr	
29		1.5x20C	M7530506015	1000 Mtr	
30		1.5x24C	M7530506074	1000 Mtr	
31		1.5x30C	M7530506016	1000 Mtr	
32		2.5x2C	M7530506017	1000 Mtr	
33		4 Quad	M7530506051	1000 Mtr	
34		6Quad	M7530506062	1000 Mtr	
35		2x25 AI	M7530506047	1000 Mtr	
36		OFC	M7530506061	1000 Mtr	
37	Batteries	80AH	M7530426005	Procurement proposal to be initiated 1 year before completion of codal life & min 10% of population for each type.	
38		40 AH	M7530426004		
39		120AH	M7530426006	Procurement proposal to be initiated 1 year before completion of codal life & min 5 nos for each type.	
40		300AH	M7530426008		
41		300AH	M7530426003		
42	Others	IPS	M7530990018	AMC with spares	
43		EI System			
44		MSDAC			
45		VHF			
46		DG Set		AMC with spares	
47		Point Machine		Min 5 Nos.	
		Point machine		Min 10 % of population	

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

48		spares			
49		LC Gate Spares		Min 10 % of population	
50	VHF	Walkie-Talkie		Min. 5 nos.	AMC with spares
		Base Stations		Min. 2 nos.	
		Antennas, PTT, DC-DC Converter etc.		Min. 5 nos	

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

Annexure XIII:**List of RDSO specification for S&T items:**

S No	Spec No	Description
01	BR 930	Plug in type Neutral Line Relay Non-AC
02	BR 931A	Plug in type Neutral Line Relay ACI
03	BR 932A	Plug in type DC Biased ACI
04	BR 933A	Plug in type DC Slow Pickup ACI
05	BR 934A	Plug in type DC Slow Release ACI
06	BR 935A	Magnetically Latched Neutral Relay
07	BR 936A	DC Polarised Magnetic Stick Line Relay
08	BR 937A	Thermal Time Element Relay
09	BR 938A	Plug in type DC Neutral Track Relay Non-ACI
10	BR 939A	Plug in type DC Neutral Track Relay ACI
11	BR 940	DC Single Wound Lamp Proving Relay
12	IRS S 46-74	D. C. Neutral Line Relay (Proved Type) For Railway Signalling
13	IRS S 60-78	AC Immunisation Characteristics of DC Neutral Tractive Armature Miniature Plug-in Type Line Relays for Railway Signalling
14	IRS S 67-85	SIG Contacts for Railway Signalling Relays
15	IRS S 61-2000	Fail Safe Electronic Time Delay Device for Railway Signalling
16	IRS S 98-2001	Single Line Tokenless Block Instrument Handle Type
17	IRS S 103-2004	Welding Material for Track Circuit Applications to Make Reliable & Permanent Electrical Bond Connection Between Two Conductors
18	IRS S 100-2002	Electric Lamp Filament Switching Unit for Colour Light Signals
19	IRS S 57-2005	Electric Lamps for Railway Signalling
20	IRS S 65-83	Choke coil for Single Rail Track Circuits On 25 KV 50 Hz AC Electrified Section
21	IRS S 24-2002	Electric Point Machine Non Trailable Type
22	IRS S 77-91	120/160/200 Way Tag Blocks for Signalling Installations
23	IRS S 21-2001	Electric Key Transmitter Rotary type
24	IRS S 19-2000	D.C. IRS Post Type Electric Signal Reverser
25	IRS S 97-2000	Glass Fiber Reinforced Nylon-6 Signalling Components
26	IRS S 76-89	PVC Insulated PVC Insulated Cables & Wires for Indian Railway Signalling
27	IRS S 35-93	PVC insulated aluminium screened cables for railway Signalling
28	IRS S 63-89	PVC insulated cable for railway signalling
29	IRS S 89-93	Track feed battery charger
30	IRS S 86-2000	Battery chargers for railway S&T installations
31	IRS S 59-77	Colour Light Signal Transformer
32	IRS S 96-2000	DC-DC Converter for Railway S&T Installations
33	IRS S 41-70	Electric lifting barrier
34	IRS S 74-89	Ferro-resonant type automatic ac voltage regulator for railway signalling installations
35	IRS S 88-2004	Low maintenance lead acid stationary secondary cells for S&T installations
36	IRS S 95-96	'SITEL' primary cells (dry, Leclanché type) for railway S&T instalations

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

37	IRS S 72-88	Transformer 230/110V - 1 KVA to 5 KVA
38	IRS S 93-96	Valve regulated (sealed) lead acid stationary battery & charger
39	IRS S 91-93	Transformer - Rectifier Set
40	IRS S 99-2001	Data logger system
41	IRS S 102-2004	Solid State Interlocking system
42	IRS S 84-92	Solar photovoltaic module
43	RDSO SPN 144-2004	Safety and reliability requirement of electronic signalling equipment
44	RDSO SPN 146-2001	Audio Frequency Track Circuit
45	RDSO SPN 147-1997	Universal fail-safe block interface (UFSBI)
46	RDSO SPN 153-2002	LED Signal Lighting Unit
47	RDSO SPN 165-2000with Amd-2	SMPS Based Integrated Power Supply System (IPS) for S&T Installations of Indian Railways
48	RDSO SPN 165-2004	SMPS Based Integrated Power Supply
49	RDSO SPN 168-2002	Thermoplastic Polyurethane moulded Rail Insulation Joints
50	RDSO SPN 169-2001	LED lit Semaphore Signal lamp
51	RDSO SPN 173-2002	Fail Safe Electronic Time Delay Device For Railway Signalling
52	RDSO SPN 175-2005	Solid State Block Proving By Axle Counter
53	RDSO SPN 176-2002	Multi section Digital Axle Counter
54	RDSO SPN 177-2003	Single section digital axle counter
55	RDSO SPN 179-2003	Train actuated warning device based on axle detectors of digital axle counter technology for level crossing gate
56	RDSO SPN 183-2004	Train protection & warning system
57	RDSO SPN 184-2004	Noncorrosive NFTC apparatus case for Railway Signalling
58	RDSO SPN 188-2004	Block panel with UFSBI / MUX-combiner

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

Annexure XIV:**Safety at workplace:****A. General Safety Precaution**

- Never use short cut operation while working with battery and other equipment.
- While connecting measuring equipment ensure that it is connected with proper setting as per specified guidelines.
- Always use the instrument on higher range of the value if not known.
- Before energization, check all the connections are proper as per circuit diagram and fully tightened.
- Before disconnection of the circuit ensure that information is to be issued to the users/party.
- Maintain properly and ensure that all the measuring instruments are in working condition.
- Never use short cut.

B. Some Hazards associated with S&T Maintenance

- Signal Post – fall from height.
- Point Machine – Caught between tongue & stock rail.
- Track Circuit – Runover
- Battery – Acid spill
- IPS / Other Electrical System – Electrical shock/Fire
- Solder Machine – Burn
- Drill/Welding Machine – Electrical Shock, physical injury.
- Cable Trenching – physical injury
- Loc Box – Snack/insects bite

Other type of hazards may also be there at the workplace, all such hazards to be identified and HIRA & JSA document to be prepared for such hazards.

c. Electrical & Electronics Equipment

The following precautions are to be taken during maintenance, and troubleshooting of Electric/Electronic equipment:

- Ensure proper earthing of equipment.
- Do not touch the electric/electronic equipment unless you are sure that there is no leakage current in the body of the equipment.
- While working on the electrical/electronic equipment always use rubber shoes, mat, gloves, or dry wooden platform to maintain the separation from the earth.
- Before switching on the equipment always ensure that the circuit is properly connected, and no person is working on the circuit.
- Never use short cut methods while rectifying a failure.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

- Before starting any maintenance, repair, always disconnect equipment from mains. Never pull the plug from the socket by holding the lead wire of mains.
- Whenever working on the mains your hands should be dry. Never touch the mains, electrical equipment with wet hands.
- If there is any high voltage capacitor in the circuit always touch or disconnect the capacitor after discharging it.
- Never experiment with equipment. Before doing any testing always have full knowledge about it and all required tools; equipment and circuits with you.

D. Monsoon Season

- Ensure provision and cleaning of drainage system in yards to avoid water accumulation in points and track circuited area.
- Ensure removal of vegetation, mud and muck from track circuited portion of the track as well as on either side of tracks and in vicinity.
- Ensure that defective/worn out glued joints, nylon insulations are replaced before onset of monsoon.
- In flood prone areas ensure that Track lead Junction Boxes are raised without infringing Schedule of Dimensions.
- Check all traction bonds in track Circuited Portion of track and attending deficiencies.
- Check and ensure proper Earthing for axle counter field equipment and Evaluator, IPS.

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

Annexure XV:**Sample Format for Caution Order (S&T):**

Shift In-charge Engineer MGR (O)		Date
Maintenance of S&T equipment will be carried as per following details -		
Equipment Category	Location / Line	Details of Equipment
Signals		
Point Machine		
Track Circuit		
Others		
Date & Time of Work: - FromTo		
Kindly arrange to issue caution order to all Loco Drivers to drive cautiously, whistling frequently, observe engineering signals and be prepared to stop if required at the site of work.		
Applicant:		SEC

TITLE	Signalling and Telecommunication system maintenance
LMI No.	LMI/OGN/OPS/SYST/061 Rev.0

Annexure XVI:**List of Records:**

S No	Description of Record	Remarks
01	Signal Failure Register	To be maintained at C/R and regularly acknowledged by S&T maintenance.
02	Relay Room entry record	To be maintained at C/R.
03	Point Burst Register	
04	Equipment outage more than 3Hrs	
05	Signal Maintenance Record	
06	Point Machine Maintenance Record	
07	Track Circuit Maintenance Record	
08	Apparatus Case (Loc Box) Maintenance Record	
09	Axle Counter Maintenance Record	
10	IPS Maintenance Record	
11	Relay Room Maintenance Record	
12	LC Barrier maintenance record	
13	Block Instrument Maintenance Record	
14	VHF maintenance Record	
15	VHF equipment issue record	With details of equipment Sr No, asset no, issue to, issue date etc.